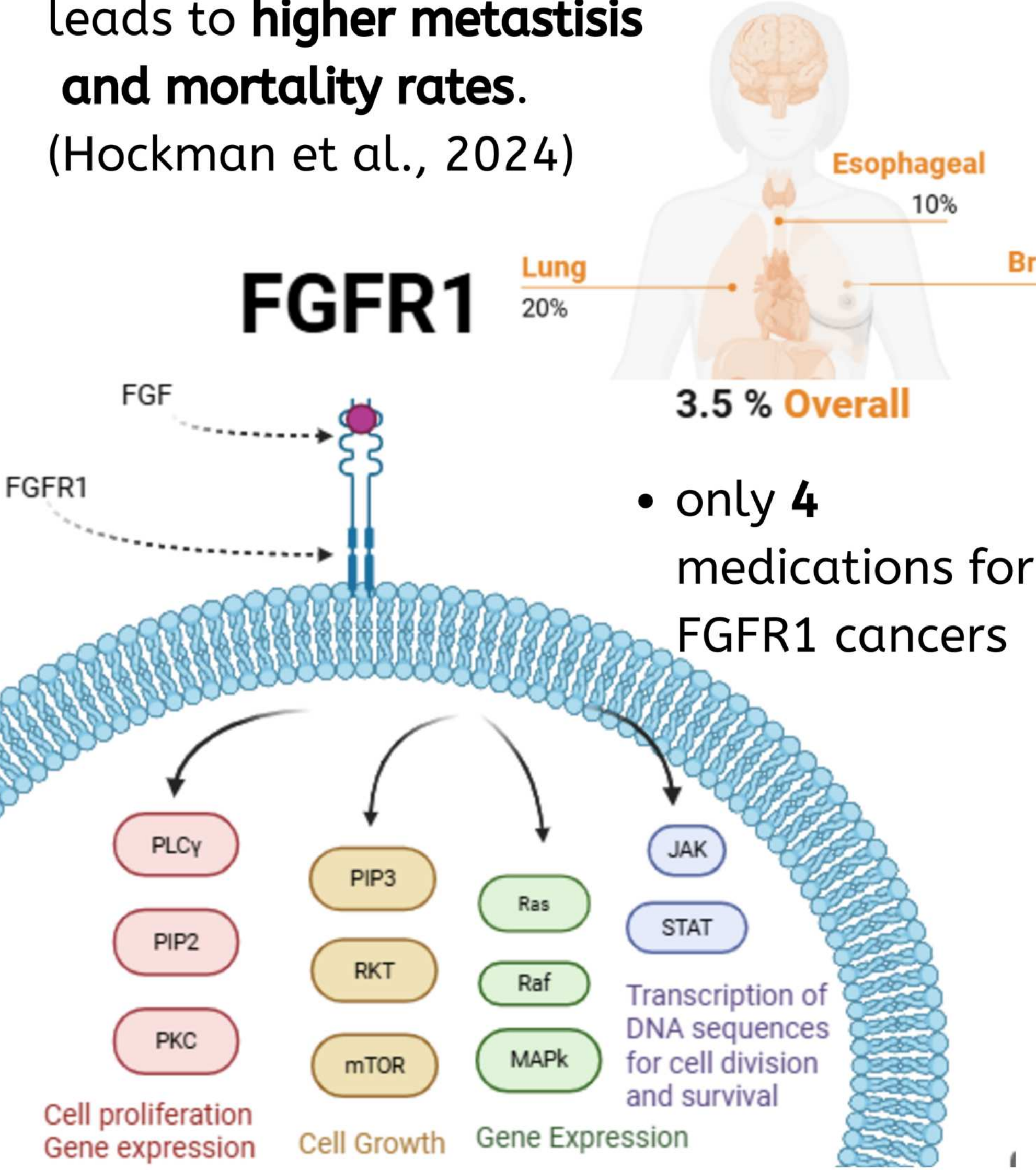


USING SAMSON MOLECULAR DOCKING TO IDENTIFY NOVEL FGFR1 INHIBITORS FROM HUMAN ENDOGENOUS METABOLITES IN-SILICO

Problem

FGFR1 is a protein that is overexpressed in types of cancer and leads to **higher metastasis and mortality rates**. (Hockman et al., 2024)



- This research uses **molecular docking** which predicts inhibition effects of compounds measured in kCal/mol
- And analyzes the effect of **Endogenous Metabolites** as safer therapeutic option (Lee, 2020)

Project framing

Research Question: What is the effect of human endogenous metabolites on FGFR1 inhibition

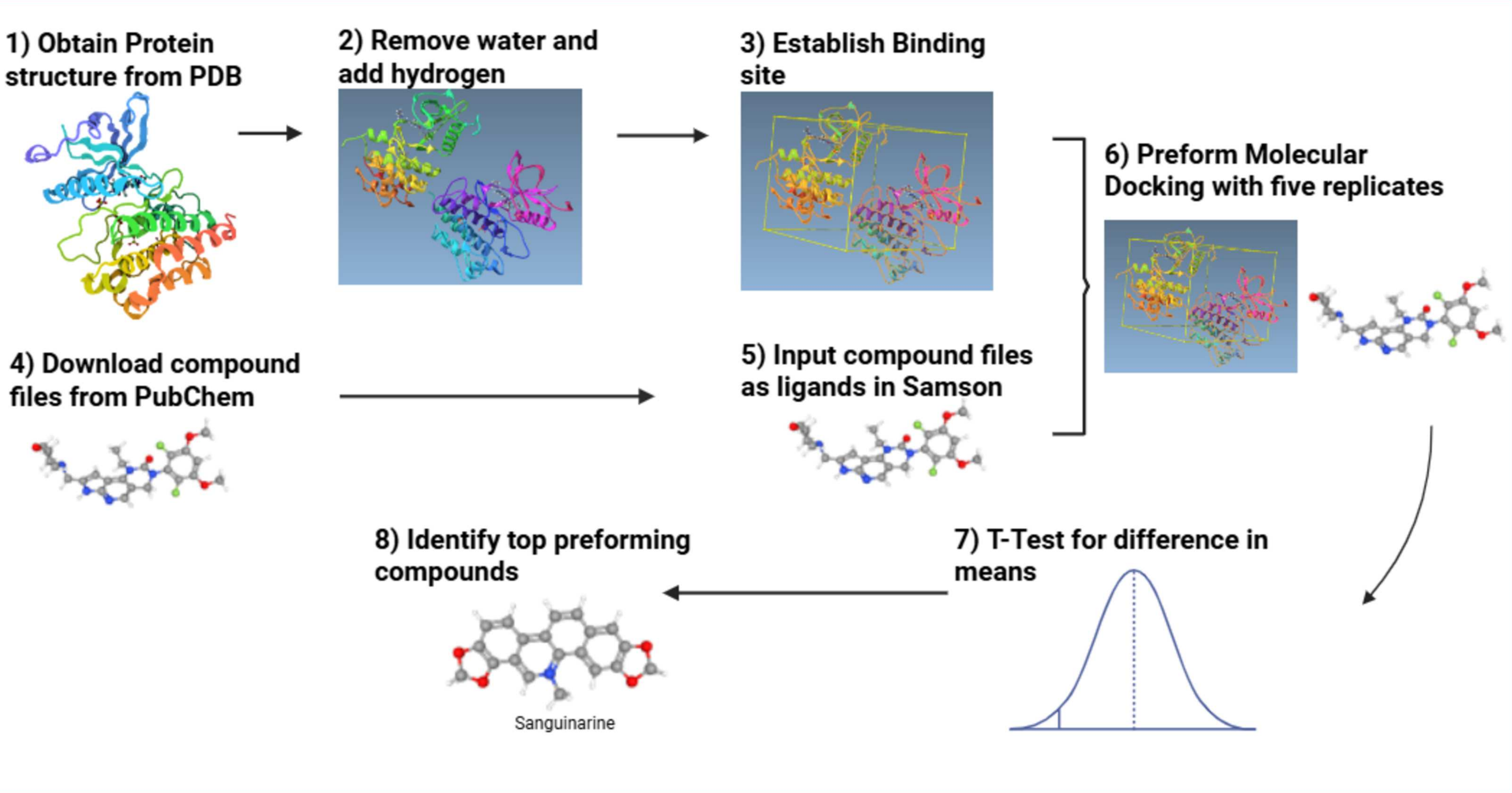
IV: the specific compound tested

DV: the binding affinity of the compound in kCal/mol

Control: the binding affinity of pemigatinib, a currently used FGFR1 cancer medication

Goal: to identify at least one compound with a similar binding affinity to FGFR1

Procedure



Results

T- tests

T test results		
Compound tested	Mean value	P value
Sanguinarine	-9.243	0.93
Bilirubin	-8.775	0.31
DHEA	-8.065	0.03

Interpretation

- Sanguinarine had highest P-value, indicating the **least difference in binding affinity** from pemigatinib
- Suggests that Sanguinarine could have a similar inhibition to Pemigatinib
- Contrasts to DHEA with low p value, shows less possible inhibitory effect than pemigatinib
- Bilirubin, had an intermediate p value compared to the two other compounds
- Greater effect than DHEA but less than sanguinarine

Implication

- Research Identified Sanguinarine as a possible novel treatment for FGFR1 cancers

Data

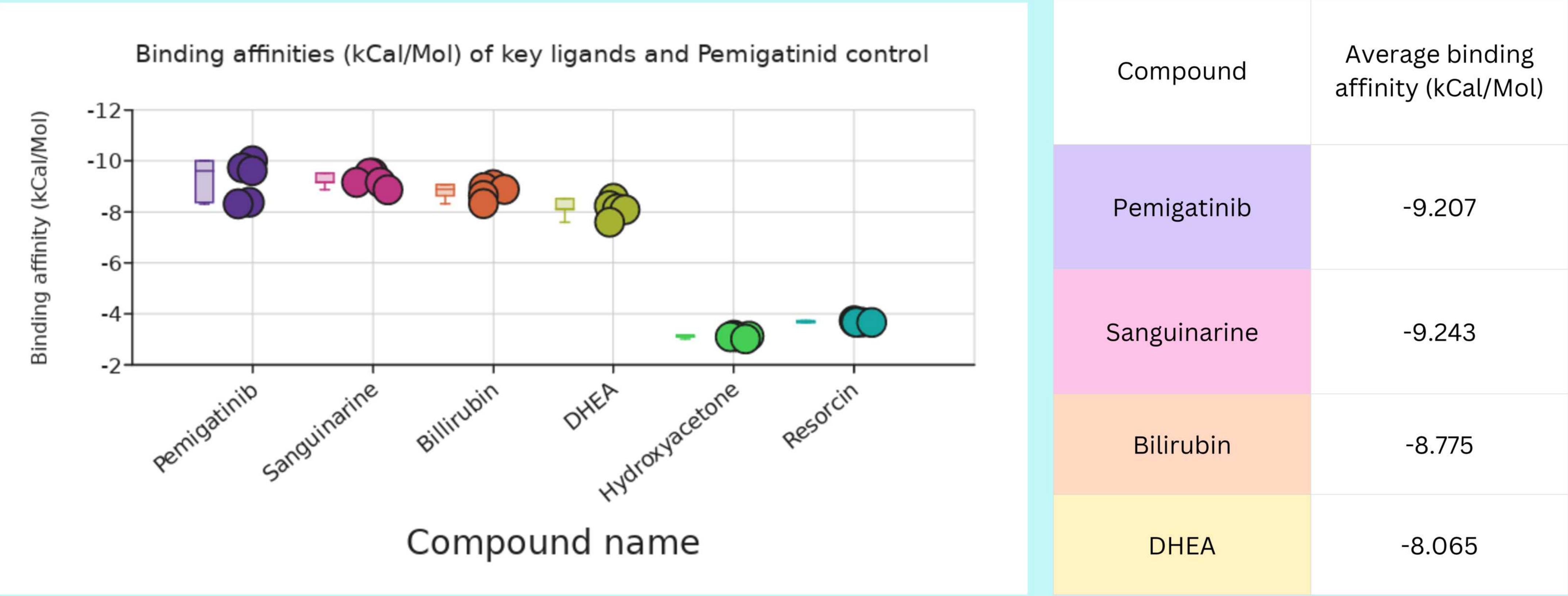


Figure 2 : graphy of the three highest binding compounds (Sanguiarine, Bilirubin, and DHEA, with control, and two low binding affinities (Resourcin and Hydroxyacetone)

Future work

- Testing compounds in cells to further evaluate their effects in FGFR1 cancers
- Preforming ADMET anlysis to evaluate the safety of these compounds as a therapeutic

Key References

Bioinformatics Insights. (2024, October 14). How to Perform Virtual Screening or Simple Docking in SAMSON? [Video]. YouTube. https://www.youtube.com/watch?v=zUVsscJ_zvk

Jamal, S. B., & Hockman, D. (2024). FGF1. Differentiation, 139, 100802–100802. <https://doi.org/10.1016/j.diff.2024.100802>.

Lee, N., Spears, M. E., Carlisle, A. E., & Kim, D. (2020). Endogenous toxic metabolites and implications in cancer therapy. *Oncogene*, 39(35), 5709–5720. <https://doi.org/10.1038/s41388-020-01395-9>

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